Complementary Spatial and Temporal Evidence for Reliable Deepfake Video Detection

Muhammad Umar Farooq, Kutub Uddin, Awais Khan, and Khalid Malik

Abstract—Deepfake detectors that read a single face crop have become very strong on face swapping, where the manipulation leaves a per-frame trace. They remain weak on face reenactment, where the frames can be individually plausible and the manipulation lives in how the face moves. We study this asymmetry directly and address it with an evidential formulation in which appearance, off-manifold and temporal evidence are combined as subjective opinions rather than averaged logits. Each view emits a Dirichlet opinion; per-view reliability enters as evidence scaling fitted on source-grouped development folds; and the opinions are combined with the direct multi-source PCR6$^+$ rule, which – unlike Dempster's rule and unlike plain PCR6 – guarantees that a view the reliability fit has discounted to zero behaves exactly as a removed view. Holding the spatial evidence fixed at a strong baseline's own values and adding only the temporal view, the system is ahead of that baseline on 39 of the 41 generator–source entries of DF40. The margin is small on face swapping, where both systems exceed 0.97 AUROC, and large on reenactment (+0.164 mean), including four generators on which both spatial views sit at chance and the temporal view alone reaches 0.94–0.99. Across ten cross-dataset benchmarks the system attains a mean video AUROC of 0.9129, the best of the ten systems compared, and the result holds within 0.009 across four detector configurations. The entire trainable footprint of the extension is 0.92M parameters, because the spatial views are frozen. A family-resolved ablation shows the three views dividing the work rather than duplicating it: the temporal view carries reenactment, the off-manifold view contributes on every evaluation group, and the appearance view earns its place on swap and in-the-wild content.

Index Terms—Deepfake detection, face reenactment, evidential deep learning, subjective logic, information fusion, conflict redistribution, uncertainty quantification, video forensics.

I. Introduction

FACE manipulation methods have separated into two families that make very different demands on a detector. Face swapping transplants an identity into a target video and, in doing so, leaves per-frame evidence: blending seams, colour and resolution mismatch, warping near the mask boundary. Face reenactment instead drives a target face with a source performance. The identity is untouched, each frame can be a photometrically clean rendering of a real person, and what is wrong is the motion – expression dynamics, head-pose coupling, the timing of blinks and mouth closure.

Most recent detectors are trained and evaluated in a way that rewards the first family. They consume one aligned crop, they are scored on datasets dominated by swaps, and they have become very accurate there. Our own conference system [1] is no exception. The consequence is visible when a detector meets a reenactment generator that renders cleanly: the per-frame evidence simply is not present, and a spatial detector has nothing to read.

This paper starts from that asymmetry and treats it as an evidence problem rather than an architecture problem. We keep the spatial evidence of a strong baseline fixed, add a temporal view that sees motion, and ask how the three bodies of evidence should be combined when they are of unequal and input-dependent reliability, and when they sometimes flatly disagree.

Three design commitments follow. First, each view emits a subjective opinion – a Dirichlet over the binary frame with an explicit vacuity mass – rather than a logit, so that "this view has little to say here" is representable and is not confused with "this view says the two classes are equally likely". Second, per-view reliability is fitted on source-grouped development folds and applied as evidence scaling, which is the discount our system implements and which differs from classical belief-mass discounting everywhere except at its endpoints. Third, the opinions are combined with the direct multi-source PCR6$^+$ rule of Dezert et al. [2]. That last choice is not cosmetic. The reliability fit routinely drives a view's weight to zero, and a zero-weighted view is exactly the vacuous source whose neutrality Dempster's rule and plain PCR6 both violate: under PCR6$^+$ removing such a view changes the fused opinion by exactly zero, whereas under PCR6 it can change it by up to 0.28.

Our contributions are:

1) A controlled demonstration of spatial–temporal complementarity. Holding two spatial views frozen at a strong baseline's own values and adding only an adapted V-JEPA 2 temporal view, the system is ahead on 39 of 41 DF40 generator–source entries. The gain is concentrated exactly where the mechanism predicts: +0.164 mean AUROC on reenactment against +0.009 on swapping (Sec. V-B).
2) An evidential fusion formulation whose operator respects the reliability it is given. We specify the opinion construction, the evidence-scaling discount and direct multi-source PCR6$^+$ fusion, and we verify the vacuous-view neutrality property algebraically on the implemented binary domain (Sec. III-E, Table X).

The authors are with the College of Innovation and Technology, University of Michigan, Flint, MI 48502 USA (e-mail: {mufarooq, kutub, mawais, drmalik}@umich.edu).

This work extends our conference paper [1], "Uncertainty-Aware Deepfake Detection via Multi-View Structural Learning".

3) A family-resolved account of what each view contributes. Removing the temporal view costs 0.151 AUROC on DF40 reenactment; removing the off-manifold view costs on every evaluation group; removing the appearance view costs on swap and in-the-wild content while helping slightly on reenactment. The three views divide the work by manipulation family rather than duplicating it, and the fitted reliability is what lets each step aside where it is uninformative (Sec. V-D).
4) A reliability and robustness account covering calibration, error ranking, selective prediction, and evaluation-time JPEG and resize corruption, with the fused vacuity established as the best of four candidate uncertainty scores and the retained conflict mass characterised as localising disagreement without resolving it (Sec. V-F).
5) Replication across four detector configurations obtained by changing the temporal seed and the spatial training recipe, with the swap/reenactment asymmetry persisting in all four (Sec. V-F).

II. Related Work

A. Generalising deepfake detectors

Cross-dataset generalisation has been pursued through blending-artifact simulation, frequency cues, and large pretrained visual backbones adapted with lightweight modules. Recent strong baselines in our comparison – GenD variants, Effort, ForAda, FSFM [3], [4], [5], [6] – differ mainly in which pretrained representation they adapt and how they regularise it. They are spatial: each consumes an aligned face crop. Our results show these systems clustering tightly on swap-dominated benchmarks and separating sharply on reenactment, which is the behaviour this paper sets out to explain and correct.

B. Temporal modelling for face forgery

Temporal approaches range from optical-flow and landmark-dynamics features to video transformers. Self-supervised video representations such as V-JEPA 2 [7] are attractive here because they are trained to predict in representation space and therefore encode motion regularities without requiring forgery labels. We adapt such a backbone with low-rank adapters rather than full fine-tuning, which keeps the temporal view a bounded addition to an existing spatial system.

C. Evidential learning and multi-view fusion

Evidential deep learning [8] places a Dirichlet over the class simplex so that a prediction carries an explicit vacuity. In multi-view settings the question becomes how to combine opinions. Trusted multi-view classification uses Dempster's rule; RCML [9] argues that Dempster's rule understates uncertainty under conflict and instead aggregates opinions by averaging, proving that fused uncertainty then increases for conflictive instances; Discounted Belief Fusion [10] measures a pairwise degree of conflict and converts it into per-modality discounts before an order-invariant fusion. Our setting differs in that reliability is fitted once on development folds rather than inferred per sample, and the operator is chosen for a property – neutrality of a discounted-away view – that matters precisely because fitted reliabilities reach zero.

D. The closest baseline

The system we compare against throughout is DiCoME [11], which is itself a multi-view evidential detector. It attributes the failure of single-view detectors to a semantic masking effect, in which dominant identity and semantic features overwhelm subtle artifact cues in an entangled representation. Its response is to divide the representation into complementary views, purify them, map each to a Dirichlet and aggregate the resulting subjective opinions with Dempster–Shafer fusion, using the epistemic conflict between views as an uncertainty signal. Our system shares that starting point – views as opinions, conflict as information – and differs in three respects: the views are not all spatial, reliability is a fitted quantity rather than an implicit one, and the combination rule is chosen for its behaviour when a view is discounted away. We hold DiCoME's two spatial views fixed and add to them, which makes the comparison in Sec. V-B a controlled one.

E. Conflict redistribution

Dempster's rule normalises conflicting mass away, which is what produces its well-known counter-intuitive behaviour under strong disagreement. Proportional conflict redistribution rules instead return conflicting mass to the sources that produced it [12]. PCR6 [13] is the standard multi-source form; Dezert et al. [2] show that PCR5 and PCR6 can still assign conflicting mass to an ignorant source, and restore neutrality of the vacuous belief assignment through binary keeping indexes, giving PCR5$^+$ and PCR6$^+$. We use the direct multi-source PCR6$^+$ rule and credit it as theirs; our contribution here is its application to a reliability-discounted three-view detector and the demonstration that the property it restores is one our system actually exercises.

F. Relation to the conference version

This paper extends [1], which combined visual, semantic-predicate and structural streams under a Dirichlet evidential framework with confidence-modulated fusion, inter-branch disagreement calibration and per-stream auxiliary supervision, evaluated on five cross-dataset benchmarks. Table I states what is retained, modified, replaced, removed and new. In short: the evidential treatment of heterogeneous streams is retained; the semantic-predicate stream is removed and the structural stream replaced by a single off-manifold deviation score; a temporal view is added; the two fusion mechanisms are replaced by a fitted reliability discount and a conflict-redistribution rule;

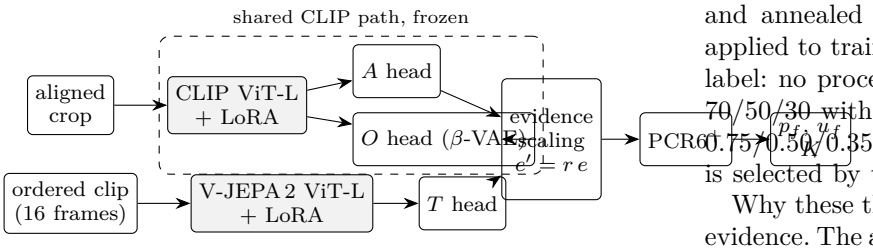


Fig. 1. The three-view evidential system. The appearance and off-manifold heads share one CLIP feature path and are frozen at the baseline's values; only the temporal view is added and adapted. Reliability enters as evidence scaling before fusion.

and the evaluation is extended from five benchmarks to ten plus 41 DF40 generator–source entries. The reenactment failure this paper addresses is not visible at five swap-dominated benchmarks, which is why it was not identified in the conference version.

III. Method

A. Overview

The system scores a video with three evidential views (Fig. 1). Two are spatial and operate on aligned face crops: an appearance view A and an off-manifold view O. One is temporal: an adapted video encoder T that consumes an ordered clip. Each view emits a binary Dirichlet opinion per frame or clip. Opinions are averaged per video, discounted by a per-view reliability fitted on development folds, and combined with direct multi-source PCR6^{+}. The fused opinion yields a probability, a vacuity, and a retained conflict mass.

B. Views

Appearance (A). A CLIP ViT-L/14 encoder adapted with LoRA [14] (rank 8, $\alpha = 16$, dropout 0.1, on the query and value projections), followed by a projection to a 64-dimensional feature and an evidential head.

Off-manifold (O). A β-VAE [15] head (32-dimensional latent, $\beta_{\mathrm{KLD}} = 2.0$) attached to the same CLIP feature path, trained with a reconstruction and KL objective alongside the classification loss. It scores how far a crop lies from the manifold of genuine faces. Because A and O read one feature path, their evidence is not independent – a fact that matters for both the fusion operator and the ablations.

Temporal (T). A V-JEPA 2 ViT-L encoder consuming 16 frames sampled uniformly and in order over the stored crops of a video. All backbone weights are frozen; LoRA adapters (rank 8, $\alpha = 16$, dropout 0.1) are inserted in the attention query and value projections of every layer, and an evidential head (LayerNorm $\rightarrow$ 128 GELU $\rightarrow$ 2, softplus) is trained on mean-pooled tokens. Training uses FF++ c23 training videos only, class-balanced sampling, AdamW (weight decay 0.01, cosine schedule to 10^{-6}, bf16), 10 epochs, batch 32, learning rates 2×10^{-4} for adapters and 2×10^{-3} for the head, with the evidential loss and annealed KL. A shared-processing augmentation is applied to training clips only, drawn independently of the label: no processing with probability 0.40, JPEG quality 70/50/30 with probability 0.10 each, and down–up resize 0.75/0.50/0.35 with probability 0.10 each. The checkpoint is selected by the development objective of Sec. IV.

Why these three. The views are chosen to read different evidence. The appearance view reads the local photometric record of a manipulation – blending seams, colour and resolution mismatch, texture inconsistency – which is what a swap leaves behind. The off-manifold view asks a different question of the same features, namely how far the crop lies from the manifold of genuine faces, which does not require the artifact to be nameable. The temporal view reads motion, which is the only channel in which a cleanly rendered reenactment can betray itself. Section V-D shows that this division of labour is real and that each view carries a family the others do not.

What is not used. The spatial checkpoint also contains a fourth evidential head, a face-prior branch distilled from a face-representation model. We evaluated it as a candidate view under the same operator and fitting procedure; the reliability fit drove its weight to zero in all five folds, whether it was added to the deployed set or substituted for the appearance view, so the fused system was numerically unchanged by its presence. It is therefore excluded and the fusion is fixed at three views. The spatial views themselves are frozen: they are warm-started from our trained baseline and not updated. The system's only new trainable component relative to that baseline is the temporal view.

Freezing the spatial views is a deliberate choice, not a convenience. Retraining the spatial encoder alongside a new temporal branch would confound two changes, and any gain could then be attributed to either. By holding the spatial evidence at the baseline's own values we make Sec. V-B a controlled contrast in which the only difference between the two systems is the presence of the temporal view and the rule used to combine the evidence. The cost of this choice is that the system inherits whatever the baseline's spatial views cannot see; the benefit is that the attribution of the result is unambiguous.

C. Opinions

We work on the binary frame $\Theta = \{\mathrm{fake}, \mathrm{genuine}\}$ with a uniform base rate and $W = 2$. A view producing non-negative evidence e_k for class k defines a Dirichlet with $\alpha_k = e_k + 1$ and $S = \sum_k \alpha_k$, and the corresponding opinion

$$b_k = \frac{e_k}{S}, \qquad u = \frac{W}{S}, \qquad p_{\mathrm{fake}} = b_{\mathrm{fake}} + \frac{u}{2}, \tag{1}$$

with $b_{\mathrm{fake}} + b_{\mathrm{genuine}} + u = 1$. The projected probability p_{fake} is the Dirichlet mean. A view with no evidence has $u = 1$ and is vacuous.

D. Reliability as evidence scaling

Per-view reliability $r_v \in [0, 1]$ is applied in the evidence domain,

$$e'_{v,k} = r_v\, e_{v,k}, \tag{2}$$

TABLE I
Relation to the conference version [1]. The extension keeps the evidential treatment of heterogeneous streams and replaces everything about how those streams are constituted and combined.

Conference component	Status	This paper	Why
Visual stream: CLIP with layer-norm tuning	modified	appearance view A: CLIP ViT-L with LoRA adapters on the attention projections	adapters leave the backbone untouched, so the view can be frozen and reused as a fixed body of evidence
Semantic stream: differentiable predicates over facial signals (expression–action-unit alignment, pose–gaze consistency, geometric symmetry)	removed	—	the predicates are per-frame and appearance-derived; they do not address the reenactment failure this paper targets
Structural stream: class-dependent dependency modelling, retained as interpretable structure	replaced	off-manifold view O: a β-VAE head on the same CLIP path, scoring distance from the genuine-face manifold	a single scalar deviation score proved to be the more reliable of the two spatial signals, contributing on every evaluation group (Table VIII)
—	new	temporal view T: V-JEPA 2 ViT-L with LoRA adapters and an evidential head on ordered clips	the conference system had no access to motion, which is the only channel in which a cleanly rendered reenactment differs from a real video
Confidence-modulated evidence fusion	replaced	reliability as evidence scaling, $e' = r\,e$, fitted per fold on source-grouped development folds	a fitted per-view reliability is auditable and can be reported per fold; it also makes zero weights explicit rather than implicit
Inter-branch disagreement calibration	replaced	direct multi-source PCR6$^+$ with retained conflict mass K	conflict is redistributed to the views that produced it and reported, instead of being converted into a calibration penalty during training
Per-stream auxiliary supervision	retained	per-view evidential supervision in the spatial checkpoint	unchanged from the conference recipe
Evaluation: five cross-dataset benchmarks	extended	ten cross-dataset benchmarks, 41 DF40 generator–source entries, and two evaluation-time processing conditions	the swap/reenactment asymmetry is invisible at five swap-dominated benchmarks

and the discounted opinion is recovered through (1). This is not classical belief-mass discounting, $m'(A) = r\,m(A)$ for $A \neq \Theta$ with the remainder to Θ; the two agree only at $r = 1$ (the identity) and at $r = 0$ (both give the vacuous opinion), and differ at intermediate r. We state this explicitly because the distinction is easy to lose and because the $r = 0$ endpoint is what makes the operator choice in Sec. III-E consequential.

The weights are fitted by grid search over $r_v \in \{0, 0.25, 0.5, 0.75, 1\}$ with $\max_v r_v = 1$, maximising the development objective of Sec. IV on the fitting fold, separately for every arm and every fold. No weight is ever transferred between arms.

E. Fusion by direct multi-source PCR6$^+$

Let $X_j = (X_{j1}, \ldots, X_{jS})$ range over tuples of focal elements, one per view, with $\pi_j = \prod_i m_i(X_{ji})$ and $\pi_j(\emptyset)$ denoting a tuple whose intersection is empty. The direct multi-source PCR6 rule [13] sets $m(\emptyset) = 0$ and, for $A \neq \emptyset$,

$$m_{\mathrm{PCR6}}(A) = m_{\mathrm{Conj}}(A) + \sum_{j\,:\,A\in X_j,\ \pi_j(\emptyset)} \frac{\big(\sum_{i:X_{ji}=A} m_i(X_{ji})\big)\,\pi_j(\emptyset)}{\sum_{X\in X_j}\sum_{i:X_{ji}=X} m_i(X_{ji})}. \tag{3}$$

PCR6$^+$ [2] modifies (3) with binary keeping indexes $\kappa_j(\cdot)$, which discard focal elements that contain every other element implied in that partial conflict:

$$m_{PCR6^+}(A) = m_{\mathrm{Conj}}(A) + \sum_{j\,:\,A\in X_j,\ \pi_j(\emptyset)} \frac{\kappa_j(A)\big(\sum_{i:X_{ji}=A} m_i\big)\,\pi_j(\emptyset)}{\sum_{X\in X_j}\kappa_j(X)\sum_{i:X_{ji}=X} m_i}. \tag{4}$$

Equation (4) reduces to (3) when every keeping index equals one, and for $S = 2$ both coincide with PCR5.

On our binary frame the indexes take a simple form. A conflicting tuple must contain both singletons, so its distinct focal elements are {fake}, {genuine} and, if any view is partly ignorant, Θ. Neither singleton contains the other, so both keep index 1; Θ contains both, so $\kappa_j(\Theta) = 0$. On this domain PCR6$^+$ is exactly PCR6 with conflicting mass never redistributed to Θ. We compute the indexes from their definition rather than assuming this, and verify the consequence: a view discounted to $r = 0$ leaves the fused opinion unchanged to 0 under PCR6$^+$, while under PCR6 the same removal moves it by up to 0.28 (Table X). Since the fitted reliabilities do reach zero in most folds, this is a property the system exercises rather than a theoretical nicety.

Algorithm 1 Inference for one video

Require: aligned crops $\{x_i\}$, ordered clip c, fitted reliabilities r_T, r_A, r_O, threshold τ

$\omega_A \leftarrow \text{mean}_i \,\text{opinion}(\text{head}_A(f_{\text{CLIP}}(x_i)))$
$\omega_O \leftarrow \text{mean}_i \,\text{opinion}(\text{head}_O(f_{\text{CLIP}}(x_i)))$
$\omega_T \leftarrow \text{opinion}(\text{head}_T(f_{\text{V-JEPA}}(c)))$
for all $v \in \{T, A, O\}$ **do** ▷ reliability as evidence scaling
    $e_v \leftarrow$ evidence of ω_v; $\omega'_v \leftarrow \text{opinion}(r_v\, e_v)$
end for
$m_{\text{Conj}}, \{\pi_j(\emptyset)\} \leftarrow$ conjunctive combination of $\omega'_T, \omega'_A, \omega'_O$
$K \leftarrow \sum_j \pi_j(\emptyset)$ ▷ retained for reporting
for all conflicting tuples X_j **do**
    compute keeping indexes $\kappa_j(\cdot)$ by (4); redistribute $\pi_j(\emptyset)$
end for
$p_f \leftarrow b_{\text{fake}} + u_f/2$
return $\hat{y} = \mathbb{1}[p_f \geq \tau], \quad p_f, \quad u_f, \quad K$

F. Aggregation, outputs and decision

Per-frame opinions are averaged per video before fusion; the temporal view already emits one opinion per clip. Fusion is applied to the video-level opinions. The system outputs the fused probability p_f, the fused vacuity u_f, and the conflicting mass K accumulated before redistribution, which we also decompose by the pair of views that produced it. The decision threshold is the equal-error-rate point on the FF++ validation split, computed per arm on its own scores and applied unchanged to every evaluation set. Labels are never an inference input.

IV. Experimental Protocol

A. Training and development data

All trainable components are trained on FaceForensics++ (c23) [16] training videos only. Development uses three held-out mixes – VALmix, VALmix-v2 and the FF++ validation split – combined into a development pool that is used for two purposes and no others: selecting checkpoints and fitting the reliability weights.

The development objective is a macro AUROC over three manipulation-type cells (swap, reenactment, in-the-wild), computed over five source-grouped folds produced by GroupShuffleSplit on the source-video identity with seeds 0–4. Grouping by source keeps every manipulated copy of a source video, and its processed variants, in the same fold. These five folds are fitting folds, not detector seeds; they quantify sampling and weight-fitting variability conditional on the trained encoders, not encoder-training variability.

B. Evaluation suites

We report two suites. The first is ten cross-dataset benchmarks: CDF-v2 and CDF-v3 [17], DFD [18], DFDC [19], DFDCP [20], DeepfakeEval-2024 [21], FFIW [22], KoDF [23], IDForge [24] and DFDM [25]. The second is DF40 [26] cross-manipulation, comprising 41 generator–source entries: ten face-swap and eleven reenactment generators, each in a Celeb-DF-sourced and a FaceForensics++-sourced version, plus DeepFaceLab. Each entry is scored against the genuine pool DF40 ships for its source, so a generator's AUROC is never computed against another source's real videos. We report a family-balanced mean, which weights each of the four family×source cells equally, and a generator-macro mean, which weights each of the 41 entries equally; the two differ because the Celeb-DF-sourced cells contain more generators.

Processing robustness is evaluated by applying JPEG quality 40 and down–up resize 0.60 to the stored crops at evaluation time. These corruptions never appear in any fitting or selection step.

C. Exposure

CDF-v2 and DeepfakeEval-2024 informed development through diagnosis, and DF40 informed operator design in preceding studies. We therefore describe both suites as exposed benchmark evaluation rather than untouched confirmation, and we do not claim held-out generalisation from them. What the DF40 comparison does establish is a controlled contrast: both systems see identical videos with identical aggregation, and they differ only in the presence of the temporal view and the fusion rule.

D. Metrics

All detection numbers are video-level AUROC in $[0, 1]$. ECE uses 15 equal-width bins. Error-detection AUROC ranks the system's own mistakes at the frozen threshold using an uncertainty score, with higher indicating a more suspect sample. AURC and E-AURC follow the standard risk–coverage convention [27], with E-AURC the excess over the optimal ranking. Paired intervals are computed with a source-grouped bootstrap, 2,000 draws, resampling source groups together with all of their linked manipulated and processed copies; a difference must exceed a materiality floor of 10^{-4} AUROC to be treated as non-zero, so that two arms differing only by floating-point noise cannot produce a significant interval.

E. Comparators

Every comparator in Table II was reproduced locally and scored on the same videos with the same aggregation; no value is copied from a published table. The DF40 comparison uses our own trained instance of the baseline, replayed through its own fusion.

V. Results

A. Cross-dataset generalisation

Table II reports the ten benchmarks. The proposed system attains the best mean video AUROC, 0.9129, and leads on CDF-v3 (0.9639), KoDF (0.9600) and

DeepfakeEval-2024 (0.7235). It does not lead everywhere: DFDC, DFDCP and DFD are won by spatial baselines. The pattern is consistent with the paper's thesis rather than incidental to it – the sets where the system leads are those with substantial reenactment or talking-head content, and the sets where it trails are frame-artifact-dominated benchmarks on which strong spatial detectors are already near their ceiling.

B. Cross-manipulation: where the temporal view pays

Table III gives the DF40 summary and Table IV the per-entry detail. The proposed system is ahead of the baseline on 39 of the 41 entries. The two families behave very differently. On face swapping the mean margin is +0.009 and both systems exceed 0.97; on reenactment the mean margin is +0.164, and the system is ahead on all 22 entries.

The per-entry table and Fig. 2 show why. On DaNet, MCNet, SadTalker and TPSM in the Celeb-DF-sourced set, both spatial views sit at or below 0.62 AUROC – at chance – while the temporal view alone reaches 0.94–0.99. These are precisely the generators that render individually plausible frames and betray themselves in motion, and they are the shaded band of Fig. 3. A two-view spatial system has nothing to combine on them, which is why the baseline falls to 0.48–0.55 there. Because our spatial views are frozen at the baseline's own values, this gap is attributable to the temporal view and the fusion, not to a better spatial encoder.

We stress the symmetric observation: the system does not trade one family for the other. It is ahead on 17 of 19 swap entries as well, where the margin is small only because both systems are close to the ceiling.

C. What each view sees on its own

Before combining anything, Table V reports each view scored alone. Two patterns organise everything that follows. On the swap-dominated benchmarks the off-manifold view is the strongest single view and the temporal view is the weakest, by a wide margin: on CDF-v2 the temporal view reaches 0.7383 against the off-manifold view's 0.9736. On the reenactment-heavy benchmarks the ordering reverses: on KoDF the temporal view reaches 0.9656 against 0.8494 and 0.8548 for the two spatial views.

The last two rows sharpen this by isolating reenactment inside sets that are otherwise dominated by other manipulations. On the reenactment subset of CDF-v3 the temporal view scores 0.9741 while the best spatial view manages 0.8416; on KoDF's first-order-motion subset the gap is 0.9887 against 0.7306. These two slices are the cleanest statement of the problem this paper addresses: the same spatial evidence that is close to saturated on swapping is more than 0.25 AUROC worse on reenactment driven by the same generator family.

Table VI shows what fusion does with those slices. The fused system tracks the temporal view rather than the spatial majority – 0.9681 on CDF-v3 reenactment against 0.8430 for the two spatial views combined – which is the behaviour a reliability-weighted evidential combination should produce and a majority vote would not. The fused score sits slightly below the temporal view alone on these two slices, which is the price of retaining spatial evidence that is uninformative here; Sec. V-D quantifies that trade across all content.

D. View necessity

Table VII fits and scores every view subset identically under the proposed operator, and Table VIII reports what removing each view costs on each kind of evaluation content.

The temporal view is necessary, and decisively so on reenactment. Removing it costs 0.094 on the reenactment-heavy benchmarks and 0.151 on the DF40 reenactment family – an order of magnitude larger than any other view's effect anywhere in the table. It also costs on in-the-wild content (−0.020) and on DF40 swaps (−0.008). The one place it does not help is the swap-dominated benchmarks (+0.010), where per-frame evidence is abundant and motion adds little.

The off-manifold view is necessary everywhere. It is the only view whose removal hurts every group, from −0.005 on DF40 swaps to −0.015 on the swap-dominated benchmarks. It is the system's general-purpose spatial evidence.

The appearance view is specialised rather than redundant. Removing it costs on four of the five groups – swap-dominated sets, in-the-wild sets, DF40 swaps – and helps only on reenactment, where per-frame appearance is by construction uninformative and an insistent spatial opinion is a liability. This is precisely the behaviour the reliability discount is there to manage: the fitted weight is non-zero on the folds where appearance evidence is useful and is driven to zero on those where it is not, so the view participates when it has something to say and steps aside when it does not.

Table IX shows the fitted weights themselves. The temporal and off-manifold views are retained on every fold; the appearance view is switched off on three of the five. This is not an artefact to be explained away: it is the discount doing the job it was introduced for, and it is why the operator's behaviour on a zero-weighted view matters in practice rather than only in principle.

The aggregate ladder in Table VII should be read in that light. $T{+}O$ and the full system differ by less than 0.003 on any summary, because the appearance view's gains on swap content and its losses on reenactment partly cancel in a mean taken across both. The family-resolved view in Table VIII is the informative one, and it shows three views doing three different jobs.

E. The fusion operator

Table X varies only the combination rule, holding the view opinions and the fitted reliabilities fixed. Detection differences among the rules are small: the spread across

TABLE II
Cross-dataset generalisation, video-level AUROC. All systems are evaluated locally on the same videos with the same aggregation; no number is taken from a published table. Our rows are means over the five reliability-fitting folds. Best per column in bold.

Method	CDF-v2	CDF-v3	DFD	DFDC	DFDCP	DFEval24	FFIW	KoDF	IDForge	DFDM	Mean
DISCERN (conf. version)	0.9509	0.8954	0.9431	0.8879	0.9064	0.6869	0.9036	0.8198	0.9723	0.9929	0.8959
GenD-CLIP [3]	0.9534	0.8458	0.9350	0.8708	0.9162	0.6831	0.9278	0.8329	0.9839	0.9959	0.8945
DiCoME [11]	0.9693	0.8691	0.9343	0.8818	0.8755	0.6769	0.9284	0.8509	0.9598	0.9909	0.8937
FSFM [6]	0.9554	0.8751	0.9219	0.8844	0.9061	0.6817	0.9168	0.8467	0.9566	0.9790	0.8924
GenD-DINO [3]	0.9177	0.8172	0.9165	0.8470	0.9114	0.6536	0.9450	0.8984	0.9860	0.9903	0.8883
GenD-PE [3]	0.9548	0.8952	0.9203	0.8158	0.8860	0.6563	0.9330	0.8189	0.9808	0.9815	0.8842
Effort [4]	0.9312	0.7975	0.9276	0.8536	0.8617	0.6769	0.9252	0.8725	0.9583	0.9729	0.8777
ForAda [5]	0.9544	0.7494	0.9395	0.8726	0.9289	0.6227	0.9055	0.8739	0.9101	0.9713	0.8728
FSFM-Adapter [6]	0.8689	0.8381	0.7516	0.8318	0.8486	0.6280	0.8565	0.7405	0.9077	0.8452	0.8117
Ours (PCR6$^+$-R)	0.9583	0.9639	0.9307	0.8530	0.8597	0.7235	0.9217	0.9600	0.9631	0.9954	0.9129

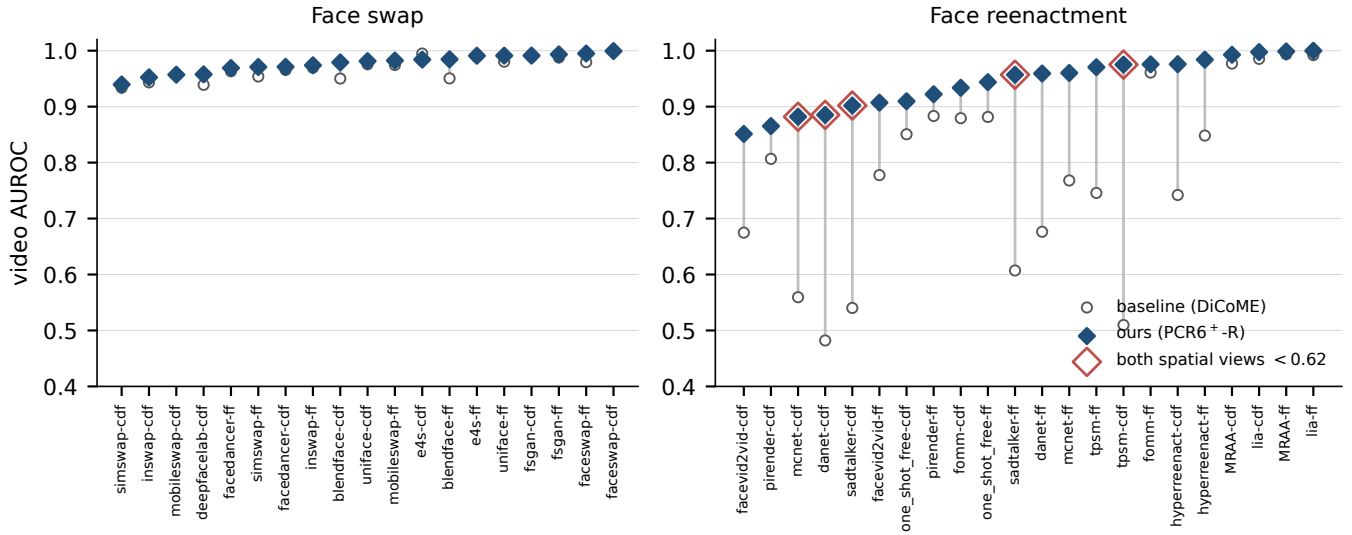


Fig. 2. Per-generator video AUROC on all 41 DF40 generator–source entries, sorted within each family by the proposed system's score. Grey stems join the two systems on the same entry. Circled markers are the reenactment generators on which both spatial views fall below 0.62 AUROC. On face swapping the two systems are close and both near the ceiling; on reenactment the baseline collapses on exactly the entries where per-frame evidence is absent.

TABLE III
Cross-manipulation on DF40. Each generator is scored against the genuine pool DF40 ships for its source. DiCoME is our own trained checkpoint replayed through its own fusion, so both systems see identical videos with identical aggregation.

Family / source	Gen.	Ours	DiCoME	Δ
Swap / CDF	10	0.9713	0.9648	+0.0065
Swap / FF++	9	0.9834	0.9720	+0.0114
Reenact. / CDF	11	0.9245	0.7280	+0.1966
Reenact. / FF++	11	0.9616	0.8304	+0.1312
Family-balanced	41	0.9602	0.8738	+0.0864
Generator-macro	41	0.9588	0.8668	+0.0920

six operators is 0.0026 on the ten-set mean. The rules are therefore not distinguished by accuracy, and we do not claim they are.

They are distinguished by the last column. Removing a view that the reliability fit has already discounted to zero changes the fused opinion by exactly 0 under PCR6$^+$ and by up to 0.28 under PCR6. Dempster's rule and the pairwise PCR6 fold share PCR6's defect. Since zero weights occur in most folds, this is the property that makes PCR6$^+$ the coherent choice for a reliability-discounted system, and it is why we adopt it despite a marginally lower mean AUROC than Dempster's rule.

Table XI gives the same comparison per benchmark. The largest spread between any two rules on any single benchmark is 0.012, and on most sets it is far smaller. This is the clearest available statement that the choice of combination rule is not what drives detection performance in this system.

Two labelling points are worth recording for reproducibility. PCR6 is not associative, so a sequential pairwise fold is a different operator from the direct multi-source rule; the two differ by up to 0.18 on admissible

TABLE IV
Per-generator cross-manipulation AUROC on all 41 DF40 generator–source entries, with the per-view scores of the proposed system. Views: T temporal, A appearance, O off-manifold. Generators where both spatial views are near chance are the ones the temporal view rescues.

Face swap						Face reenactment					
Gen.	Src	Ours	DiCoME	T	A/O	Gen.	Src	Ours	DiCoME	T	A/O
BlendFace	cdf	0.979	0.950	0.937	0.94/0.95	MRAA	cdf	0.993	0.977	0.972	0.97/0.98
DeepFaceLab	cdf	0.958	0.939	0.916	0.93/0.94	DaNet	cdf	0.885	0.482	0.952	0.46/0.53
e4s	cdf	0.984	0.995	0.806	0.99/0.99	FaceVid2Vid	cdf	0.851	0.675	0.890	0.65/0.71
FaceDancer	cdf	0.971	0.966	0.860	0.96/0.97	FOMM	cdf	0.934	0.879	0.910	0.87/0.89
FaceSwap	cdf	0.999	0.999	0.998	1.00/1.00	HyperReenact	cdf	0.976	0.742	0.986	0.73/0.77
FSGAN	cdf	0.991	0.989	0.939	0.98/0.99	LIA	cdf	0.997	0.985	0.982	0.98/0.99
InSwapper	cdf	0.952	0.943	0.887	0.94/0.94	MCNet	cdf	0.882	0.559	0.941	0.54/0.60
MobileFaceSwap	cdf	0.957	0.957	0.799	0.95/0.96	One-Shot Free	cdf	0.910	0.851	0.888	0.84/0.86
SimSwap	cdf	0.940	0.934	0.801	0.92/0.94	PIRender	cdf	0.865	0.807	0.813	0.79/0.83
UniFace	cdf	0.981	0.976	0.914	0.97/0.97	SadTalker	cdf	0.902	0.540	0.963	0.55/0.52
BlendFace	ff	0.985	0.951	0.949	0.95/0.95	TPSM	cdf	0.975	0.510	0.991	0.51/0.52
e4s	ff	0.991	0.989	0.808	0.99/0.99	MRAA	ff	0.999	0.994	0.981	0.99/0.99
FaceDancer	ff	0.969	0.963	0.833	0.96/0.96	DaNet	ff	0.959	0.676	0.973	0.68/0.69
FaceSwap	ff	0.995	0.979	0.986	0.98/0.98	FaceVid2Vid	ff	0.907	0.778	0.912	0.78/0.78
FSGAN	ff	0.993	0.988	0.945	0.99/0.99	FOMM	ff	0.976	0.961	0.936	0.96/0.96
InSwapper	ff	0.974	0.970	0.899	0.97/0.97	HyperReenact	ff	0.984	0.848	0.982	0.85/0.85
MobileFaceSwap	ff	0.982	0.974	0.916	0.97/0.97	LIA	ff	1.000	0.992	0.993	0.99/0.99
SimSwap	ff	0.971	0.954	0.858	0.96/0.95	MCNet	ff	0.960	0.768	0.968	0.77/0.77
UniFace	ff	0.991	0.980	0.923	0.98/0.98	One-Shot Free	ff	0.944	0.882	0.927	0.88/0.87
						PIRender	ff	0.922	0.883	0.884	0.89/0.87
						SadTalker	ff	0.957	0.607	0.973	0.62/0.58
						TPSM	ff	0.971	0.746	0.977	0.75/0.75

TABLE V
What each view sees on its own, video AUROC. A single view needs no combination, so these numbers are operator-independent. The last two rows isolate reenactment inside sets that are otherwise swap-dominated, and are where the temporal and spatial views separate most sharply.

Test set	real	fake	T	A	O
CDF-v2	178	340	0.7383	0.9612	0.9736
CDF-v3	178	4940	0.9263	0.8576	0.8801
DFD	363	3068	0.7967	0.9304	0.9338
DFDC	2315	2389	0.7009	0.8758	0.8808
DFDCP	230	424	0.7408	0.8698	0.8727
DFEval24	428	386	0.6732	0.6741	0.6688
FFIW	1735	1735	0.8011	0.9186	0.9354
KoDF	403	992	0.9656	0.8494	0.8548
IDForge	2987	2987	0.8899	0.9510	0.9635
DFDM	178	1718	0.9798	0.9886	0.9920
CDF-v3, reenactment only	178	3200	0.9741	0.8137	0.8416
KoDF, first-order motion	403	352	0.9887	0.7164	0.7306

TABLE VI
Reenactment isolated inside two benchmarks that are otherwise dominated by other manipulations. The fused system tracks the temporal view rather than the spatial majority.

Slice	Ours	T+O	A+O	T alone	best spatial
CDF-v3 reenactment	0.9681	0.9727	0.8430	0.9741	0.8416
KoDF FOM	0.9833	0.9860	0.7310	0.9887	0.7306

opinions and are reported as separate rows. Yager's rule is likewise non-associative and is implemented directly.

TABLE VII
View subsets, each fitted and scored identically and fused with the proposed operator; only the view set changes. Best per column in bold. These are means taken across manipulation families and partly cancel the per-family effects; Table VIII resolves them.

Views	Dev. v3.1	Reenact.	10-set	DF40
T	0.8094	0.9135	0.8213	0.9195
A+O	0.8262	0.7644	0.8959	0.8804
T+A	0.8693	0.9065	0.9003	0.9522
T+O	0.8800	0.9157	0.9121	0.9624
T+A+O (deployed)	0.8794	0.9111	0.9129	0.9602

TABLE VIII
What each view contributes, and where. Each column removes one view from the deployed set and reports the change in mean video AUROC. The views divide the work by manipulation family: the appearance view earns its place on swap and in-the-wild content, the off-manifold and temporal views on reenactment.

Evaluation group	Full	$-A$	$-O$	$-T$
Swap-dominated sets	0.9267	−0.0015	−0.0152	+0.0097
Reenactment-heavy sets	0.9620	+0.0027	−0.0092	−0.0941
In-the-wild sets	0.8226	−0.0024	−0.0083	−0.0201
DF40 face swap	0.9770	−0.0010	−0.0053	−0.0080
DF40 reenactment	0.9431	+0.0054	−0.0110	−0.1514

F. Reliability, calibration and robustness

Table XII reports calibration and selective prediction per benchmark, and the two evaluation-time corruptions. The fused vacuity is an informative error ranker, and risk falls monotonically as coverage is reduced, which is the behaviour a deployed system needs when it is allowed to abstain.

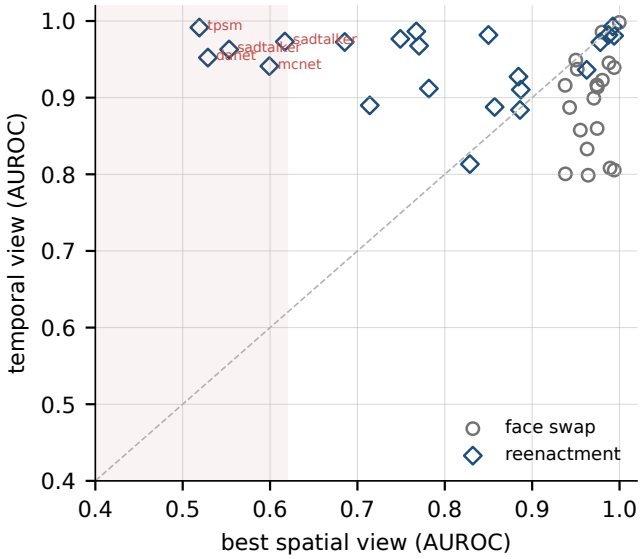


Fig. 3. Complementarity structure across the 41 DF40 entries: the temporal view's AUROC against the better of the two spatial views. Points above the diagonal are entries where motion evidence is the stronger signal. In the shaded band both spatial views are near chance, and every point there is a reenactment generator.

TABLE IX
Reliability weights fitted per fold for the proposed system, and how often each view is discounted away entirely. The temporal and off-manifold views are always retained; the appearance view is switched off on the folds where its evidence does not help, which is the behaviour Table VIII resolves by manipulation family.

View	fold 1	fold 2	fold 3	fold 4	fold 5	zero in
T	0.25	0.25	0.25	0.25	0.25	0/5
A	0.25	0.25	0	0	0	3/5
O	1	1	1	1	1	0/5

Under corruption the system degrades gracefully rather than collapsing: neither JPEG quality 40 nor a 0.60 down–up resize was seen in any fitting or selection step, and both are applied to the stored crops at evaluation only.

G. Which uncertainty score ranks errors best

The system emits three quantities that can be read as uncertainty: the fused vacuity, the retained conflict mass K, and the sharpness of the decision itself. Table XIII compares them as error rankers on every benchmark. Entropy and margin are identical by construction on a binary frame, both being strictly monotone in $|p - 0.5|$, and are reported separately only for completeness. Across the ten sets the fused vacuity ranks errors best on average (0.738), ahead of decision sharpness (0.731), with the conflict mass clearly worst (0.686). The margin between the first two is small and does not hold on every set, but the ordering is the one the evidential formulation predicts: the quantity that represents an absence of evidence is a better guide to error than the sharpness of the decision that was nevertheless taken.

TABLE X
Combination rules over the same three view opinions with the same fitted reliabilities. Only the operator changes. $\Delta_{r=0}$ is the largest change caused by removing a view the reliability fit has already discounted to zero: it is exactly zero only for the proposed rule.

Operator	Dev. v3.1	10-set	DF40	$\Delta_{r=0}$
PCR6$^+$-R (ours)	0.8794	0.9129	0.9602	0
PCR6-R	0.8793	0.9134	0.9582	0.28
PCR6-R, pairwise fold	0.8802	0.9147	0.9586	–
Yager-R	0.8787	0.9124	0.9585	–
Averaging-R [9]	0.8799	0.9141	0.9618	–
Dempster-R [28]	0.8808	0.9155	0.9591	–

That ordering is informative about what conflict does and does not measure, and Table XIV makes it concrete on development.

Conflict in this system is almost entirely temporal–spatial: K_{AO} and K_{all} are an order of magnitude smaller than K_{TS} in every case, which is what one expects when two of the three views share a feature path. More importantly, K_{TS} is high in both disagreement cases – when the temporal view is the one that is right and when it is the one that is wrong. Conflict therefore marks that the views disagree, not which of them to believe, and a score built on it inherits that limitation. We report the conflict mass because it localises disagreement, which is useful diagnostically, and we do not use it as the system's uncertainty output.

H. Robustness across detector configurations

All results so far come from a single deployed configuration. Table XV repeats the DF40 evaluation under four, obtained by changing the temporal branch's training seed and by substituting a different spatial training recipe. Two things hold across all four: the family-balanced score stays within 0.009 of the deployed value, and the swap-versus-reenactment gap persists in the same direction.

The table should be read with its limitation stated. The first two configurations share a spatial recipe and differ only in the temporal seed. The last two use a different spatial training recipe, so the four configurations differ in more than a seed, and they are not four independent draws of one training procedure. The deployed spatial recipe exists at one seed, and producing a second would require retraining, which is outside this study.

I. Cost

Because the spatial views are frozen, the entire trainable footprint of this extension is the temporal view: 786,432 LoRA parameters across 48 adapted attention projections and a 133,506-parameter evidential head, 919,938 in total, on top of a frozen V-JEPA 2 ViT-L. The fusion adds three fitted scalars. Inference cost is one additional video encoder forward pass per clip; the combination itself is closed-form and negligible.

TABLE XI
Every combination rule on every benchmark, over the same view opinions with the same fitted reliabilities. The spread across rules is small on all ten sets; the rules are separated by the property in Table X, not by accuracy.

Operator	CDF-v2	CDF-v3	DFD	DFDC	DFDCP	DFEval24	FFIW	KoDF	IDForge	DFDM	Mean
PCR6$^+$-R	0.9583	0.9639	0.9307	0.8530	0.8597	0.7235	0.9217	0.9600	0.9631	0.9954	0.9129
PCR6-R	0.9596	0.9618	0.9318	0.8553	0.8606	0.7235	0.9245	0.9583	0.9636	0.9954	0.9134
PCR6-R pairwise	0.9607	0.9613	0.9334	0.8594	0.8633	0.7241	0.9272	0.9577	0.9648	0.9954	0.9147
Yager-R	0.9602	0.9635	0.9310	0.8507	0.8576	0.7223	0.9210	0.9600	0.9624	0.9953	0.9124
Averaging-R	0.9542	0.9632	0.9330	0.8578	0.8642	0.7236	0.9241	0.9602	0.9650	0.9960	0.9141
Dempster-R	0.9598	0.9616	0.9338	0.8626	0.8656	0.7252	0.9282	0.9574	0.9656	0.9956	0.9155

TABLE XII
Calibration, selective prediction and processing robustness for the proposed system. Error-detection AUROC ranks the system's own mistakes; E-AURC is the excess area under the risk–coverage curve. JPEG and resize are applied at evaluation only; no arm was trained on them.

Set	AUROC	ECE	NLL	err-AUROC	E-AURC
CDF-v2	0.9583	0.254	0.516	0.641	0.1119
CDF-v3	0.9639	0.283	0.517	0.768	0.0547
DFD	0.9307	0.220	0.456	0.790	0.0470
FFIW	0.9217	0.099	0.405	0.699	0.0749
KoDF	0.9600	0.128	0.345	0.821	0.0199
IDForge	0.9631	0.078	0.285	0.793	0.0314
DFDM	0.9954	0.060	0.108	0.897	0.0034
CDF-v2, JPEG q40	0.8683	0.482	1.311	0.646	0.2015
CDF-v2, resize 0.6	0.9396	0.304	0.652	0.673	0.1258
DFDCP, JPEG q40	0.8170	0.413	1.210	0.663	0.1635
DFDCP, resize 0.6	0.8033	0.208	0.674	0.663	0.1366

TABLE XIII
Error-detection AUROC of each uncertainty score for the proposed system, per benchmark. Higher is better; the score ranks the system's own mistakes at the frozen threshold. Entropy and margin coincide on a binary frame, both being monotone in $|p - 0.5|$.

Test set	fused vacuity	conflict K	entropy	margin
CDF-v2	0.641	0.665	0.667	0.667
CDF-v3	0.768	0.542	0.642	0.642
DFD	0.790	0.715	0.750	0.750
DFDC	0.672	0.667	0.708	0.708
DFDCP	0.675	0.643	0.668	0.668
DFEval24	0.627	0.568	0.614	0.614
FFIW	0.699	0.754	0.785	0.785
KoDF	0.821	0.655	0.718	0.718
IDForge	0.793	0.782	0.846	0.846
DFDM	0.897	0.870	0.908	0.908
Mean	0.738	0.686	0.731	0.731

TABLE XIV
Conflict structure on the development selection folds. K_{TS} is conflict between the temporal view and a spatial view, K_{AO} between the two spatial views, K_{all} the three-way term. Conflict is concentrated on the temporal–spatial axis and is high whenever the views disagree, whichever of them is right.

Case	videos	K_{TS}	K_{AO}	K_{all}
Temporal right, both spatial wrong	362	0.4194	0.0072	0.0142
Temporal wrong, both spatial right	520	0.3740	0.0076	0.0106
All three agree and right	1512	0.0324	0.0038	0.0095
All three wrong	214	0.0439	0.0092	0.0135

TABLE XV
DF40 under four detector configurations. The first two share a spatial recipe and differ in the temporal seed; the last two are a different spatial training recipe, so the four differ in more than a seed. Family-balanced means.

Configuration	Ours	Dempster-R
warm+gate spatial, V-JEPA seed 42 (deployed)	0.9602	0.9591
warm+gate spatial, V-JEPA seed 1337	0.9651	0.9646
SBI arm B spatial seed 42, V-JEPA seed 42	0.9635	0.9643
SBI arm B spatial seed 1337, V-JEPA seed 1337	0.9690	0.9698

VI. Discussion and Limitations

What the evidence supports. The strongest claim this paper makes is the controlled one: adding a temporal view to frozen spatial evidence changes reenactment detection substantially and swap detection marginally, in a direction and magnitude that the per-generator view scores explain. The weakest link in any contrary reading is the near-chance generators, where the spatial views are demonstrably uninformative and the temporal view is demonstrably not.

What it does not support. We do not claim a novel fusion operator: PCR6$^+$ is prior work and is credited as such. We do not claim that the three views are jointly necessary on every kind of content – Table VIII shows the opposite, that each is dispensable somewhere and that the reliability fit exploits this. We do not claim held-out generalisation from the DF40 or ten-set results, because both suites are exposed.

Seeds and replication. The five folds are fitting folds, not independent detector seeds. Results are reported for one deployed detector configuration; the spatial recipe exists at a single training seed and a second spatial seed would require retraining, which is outside this study's scope.

Reliability as a fitted quantity. Reliability weights are fitted once on development and held fixed at inference. Per-sample reliability estimation, as in [10], is a natural extension and is not attempted here.

VII. Conclusion

Face swapping and face reenactment place different demands on a detector, and the gap between them is not closed by a better spatial encoder. We showed

TABLE XVI
Trainable parameters added by this work. The spatial views are frozen, so the entire trainable footprint of the extension is the temporal view's adapters and evidential head; the fusion itself has three fitted scalars.

Component	Trainable parameters
Temporal LoRA adapters (48 attention projections)	786,432
Temporal evidential head	135,506
Appearance and off-manifold views (frozen)	0
Reliability weights (three scalars, grid-fitted)	3
Total	919,938

that adding an adapted temporal view to frozen spatial evidence, and combining the three as subjective opinions under a reliability-aware conflict-redistribution rule, leaves swap performance essentially unchanged while changing reenactment performance substantially – ahead on all 22 reenactment entries of DF40 and on 39 of 41 overall. The result holds across four detector configurations and costs 0.92M trainable parameters, because the spatial evidence is reused rather than retrained. We also showed how the work divides: the temporal view carries reenactment, the off-manifold view contributes everywhere, the appearance view earns its place on swap and in-the-wild content, and the operator matters less for accuracy than for coherence with the reliability weights it is given.

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